

# Long-Range Dependency Gap in Character-Level Language Models

CTW vs. N-gram vs. GPT-2

July 22, 2026

# Research Question

## Motivation

Frans Willems' ISIT 2026 Shannon Lecture asked:

*"How much can classical finite-memory models explain about natural language — and how far are they from neural LMs?"*

**Goal:** Quantify the *long-range dependency gap* between:

- **CTW** (Context Tree Weighting) — optimal finite-memory coding up to depth  $D$
- **N-gram KT** — same KT estimator, flat backoff
- **GPT-2** (small & large) — Transformer, effective context  $\approx 5000$  chars

**Metric:** Bits per character (BPC)  $\downarrow$  — lower is better

# Experimental Setup

## Dataset: WikiText-2

- Normalized to 28-char alphabet {a--z, space, newline}
- Train: **10 M chars**
- Validation: **1 M chars**

## Models

- CTW depths  $D \in \{3, 5, 7, 10\}$
- N-gram KT orders  $\in \{4, 6, 8, 11\}$
- GPT-2 small (117 M params)
- GPT-2 large (774 M params)

## CTW key parameters

- KT estimator at every node:

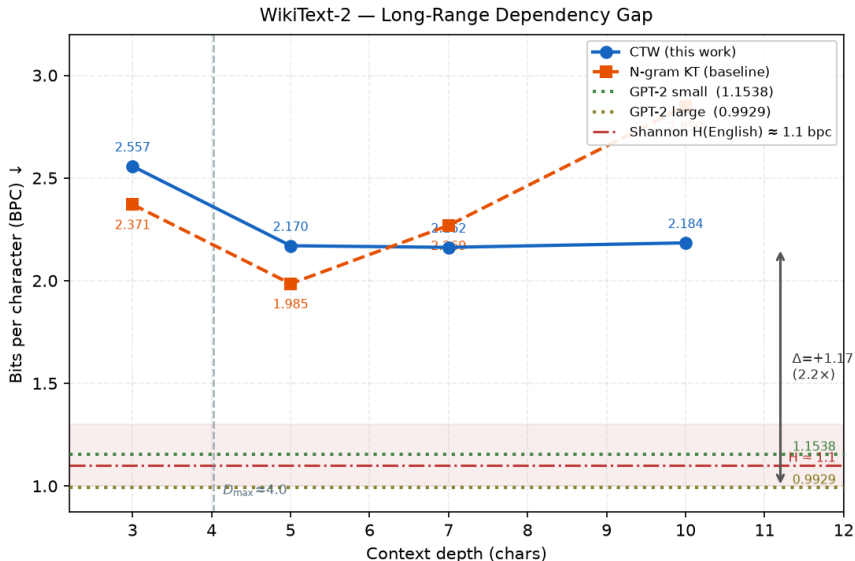
$$P_e(s) = \frac{n_s + \frac{1}{2}}{n + \frac{A}{2}}$$

- Mixing weight  $\alpha = 0.5$  (training)
- Prediction weight  $\alpha_p = 0.1$  (evaluation)
- Backoff threshold  $n_{\min} = 15$

## Second dataset

- text8 (enwiki) — replicated all experiments

# Main Results: BPC vs. Context Depth



# Surprising Finding: N-gram Beats CTW

## Why does N-gram outperform CTW?

CTW blends predictions from *all* tree levels:

$$P_{\text{pred}} = (1 - \alpha_p) P_{\text{leaf}} + \alpha_p P_{\text{parent}}$$

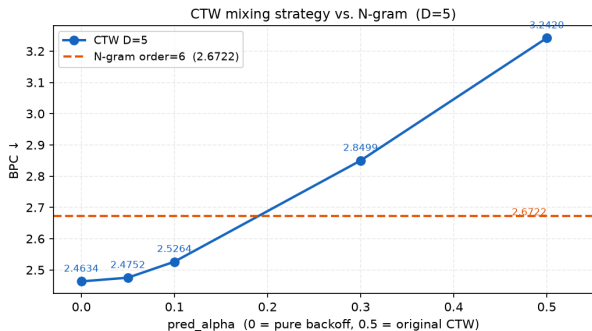
For depth  $D = 5$  with  $\alpha_p = 0.1$ :

$$0.9^5 \approx 59\% \text{ weight on leaf}$$

N-gram uses 100% of deepest available context.

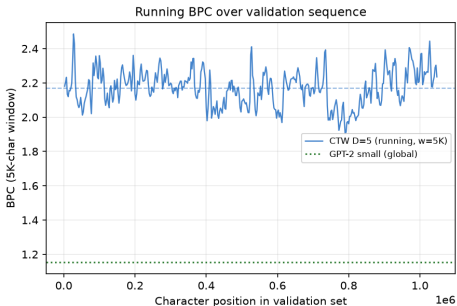
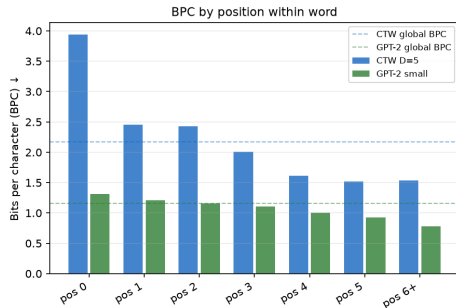
## Ablation result:

- $\alpha_p = 0.0$ : BPC = 2.463 (best CTW)
- $\alpha_p = 0.1$ : BPC = 2.526 (default)
- $\alpha_p = 0.5$ : BPC = 3.242 (original CTW)
- N-gram 6-gram: BPC = 2.672 (500K subset)



# Where Is the Gap Largest?

CTW vs. GPT-2: Where is the Gap Largest?



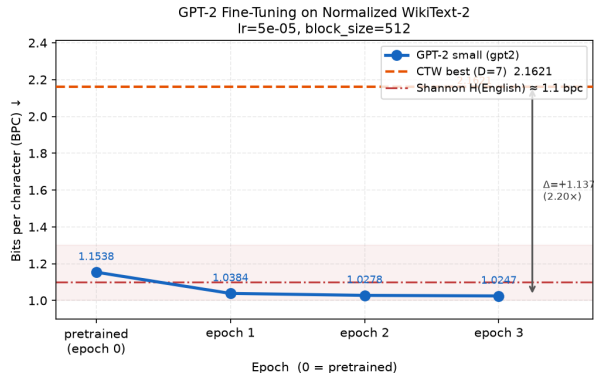
## Key finding:

The gap is  $4\times$  larger at word starts than deep within words.

Predicting the first letter of the next word requires knowing which word will appear — a semantic dependency CTW cannot resolve.

| Position       | CTW   | GPT-2 | Gap   |
|----------------|-------|-------|-------|
| 0 — word start | 3.941 | 1.313 | +2.63 |
| 1              | 2.456 | 1.207 | +1.25 |
| 2              | 2.428 | 1.155 | +1.27 |
| 5+             | 1.519 | 0.924 | +0.60 |

# Is the Comparison Fair? Fine-Tuning GPT-2



**Concern:** GPT-2 was pretrained on rich text; evaluating on 28-char normalized text may inflate the gap.

**Experiment:** Fine-tune GPT-2 small on the same 10 M normalized chars (3 epochs).

| Model            | BPC          |
|------------------|--------------|
| Pretrained       | 1.154        |
| Fine-tuned ep. 1 | 1.038        |
| Fine-tuned ep. 3 | <b>1.025</b> |
| CTW $D = 7$      | 2.162        |

⇒ Fine-tuning *widens* the gap to  $\Delta = 1.14$  bpc (2.2×).

The gap is **real**, not an artifact.

# Theoretical Saturation Depth $D_{\max}$

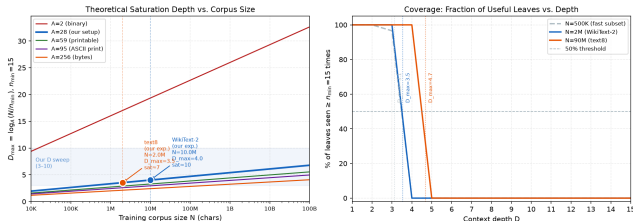
Formula:

$$D_{\max} = \log_A \left( \frac{N}{n_{\min}} \right)$$

Our setup ( $N = 10\text{M}$ ,  $A = 28$ ,  $n_{\min} = 15$ ):

$$D_{\max} \approx 4.0$$

CTW Saturation Depth:  $D_{\max} = \log_A (N/n_{\min})$



Implication:

Beyond  $D \approx 4$ , most tree leaves have  $< 15$  training samples — backoff fires everywhere.

This explains why BPC at  $D = 7$  (2.162) and  $D = 10$  (2.184) are nearly identical.

**Theory  $\Leftrightarrow$  practice:** empirical saturation matches  $D_{\max}$ .

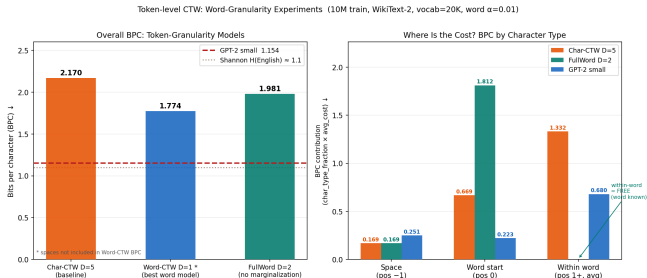


# Token-level CTW: Word-Granularity Experiments

Three variants (10 M train, vocab=20 K):

| Model              | BPC            |
|--------------------|----------------|
| Char-CTW $D = 5$   | 2.170          |
| Word-CTW $D = 1$ * | 1.774 * spaces |
| FullWord $D = 2$   | <b>1.981</b>   |
| GPT-2 small        | 1.154          |

not counted

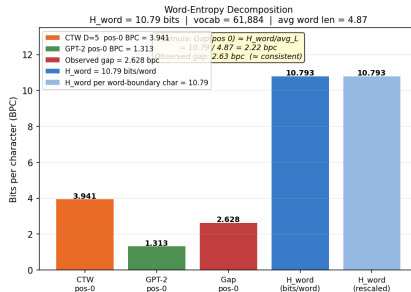
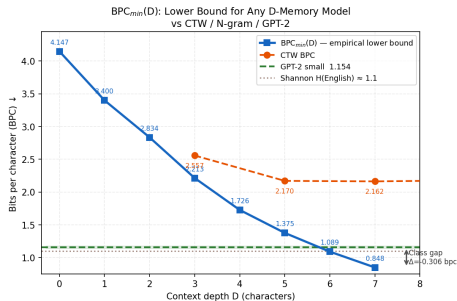


## Key findings:

- Within-word chars are free in FullWord (word identity determines all of them)
- Word-start cost dominates: **1.81** bpc vs. 0.67 for Char-CTW, 0.22 for GPT-2
- $D_{\max} \approx 1.3$  confirmed:  $D = 2$  worse<sub>9/12</sub>

# BPC<sub>min</sub>(D): Lower Bound for Any D-Memory Model

Information-Theoretic Lower Bound & Word-Entropy Decomposition  
WikiText-2, 10M train chars, 28-char alphabet



| Model              | D | BPC          | Gap to GPT-2 |
|--------------------|---|--------------|--------------|
| BPC <sub>min</sub> | 5 | 1.641        | +0.49        |
| BPC <sub>min</sub> | 7 | <b>1.205</b> | +0.05        |
| CTW                | 7 | 2.162        | +1.01        |
| GPT-2 small        | — | 1.154        | —            |

## Surprising finding:

BPC<sub>min</sub>(D=7) ≈ 1.2 is near GPT-2 — the *class gap* is only ≈ 0.05 bpc.

CTW lags BPC<sub>min</sub> by 0.96 bpc due to **data scarcity** (D<sub>max</sub> ≈ 4).

⇒ The CTW–GPT-2 gap is not a fundamental class barrier, but a **training-data problem**.

## Main Result

**CTW  $D = 7$ : 2.162 BPC** vs. **GPT-2 large: 0.993 BPC**  $\Rightarrow$  gap of **1.17 bits/char** ( $2.25\times$  more uncertainty)

- ❶ **Gap source:** concentrated at word-initial positions  
(+2.63 bpc at pos 0 vs. +0.60 bpc at pos 5+)  
 $\Rightarrow$  CTW lacks semantic/syntactic long-range context
- ❷ **Mixing vs. backoff:** pure backoff ( $\alpha_p = 0$ ) is optimal for offline prediction; original CTW mixing hurts by +0.78 bpc
- ❸ **Fair comparison:** fine-tuning GPT-2 on same data widens the gap ( $1.154 \rightarrow 1.025$  BPC), not narrows it
- ❹ **Saturation:**  $D_{\max} \approx 4$  explains why going beyond  $D = 5$  yields no improvement
- ❺ **Lower bound:**  $\text{BPC}_{\min}(D=7) \approx 1.20$  is only 0.05 bpc above GPT-2 — the class gap is negligible; the CTW–GPT-2 gap is a *data-scarcity* barrier
- ❻ **Broader context:** GPT-2 large ( $\approx 0.99$  bpc) approaches Shannon entropy ( $\approx 1.1$  bpc) — neural LMs have essentially “solved” character-level English

## Empirical

### ① PPM/PPMD baseline

Classical compression SOTA ( $\approx 2.0$  bpc).

Completes the picture:

CTW  $\rightarrow$  N-gram  $\rightarrow$  PPM  $\rightarrow$  GPT-2.

### ② Mutual information curve

$I(X_t; X_{t-d})$  vs. lag  $d$  —

shows how much long-range information exists in the corpus at each distance.

### ③ Small LSTM baseline

Fills the space between CTW and GPT-2;  
shows even a tiny neural model captures what CTW cannot.

## Theoretical

### ④ Information-theoretic lower bound ✓

$BPC_{\min}(D) = H(X_t | X_{t-D:t-1})$  computed —  
see previous slide.

**Revised claim:** gap is a *data-scarcity* barrier,  
not a class barrier ( $BPC_{\min}(7) \approx$  GPT-2).

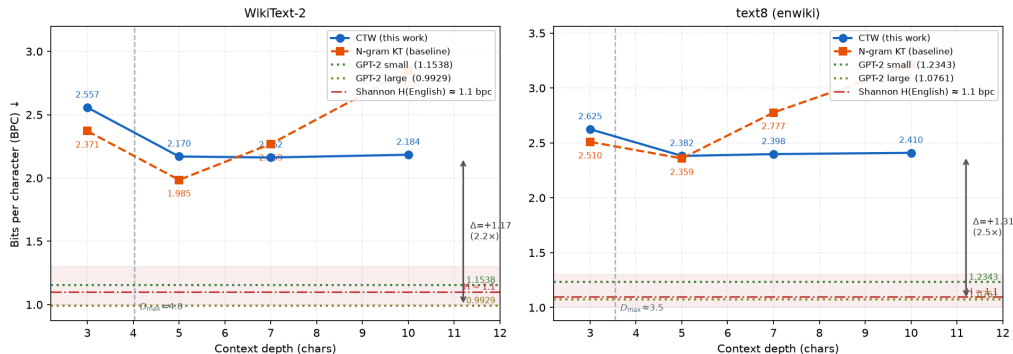
### ⑤ Word-entropy decomposition

Gap  $\approx H_{\text{word}} / \bar{L}$  at word boundaries,  
where  $H_{\text{word}}$  = word unigram entropy,  $\bar{L}$  = avg.  
word length.

Turns an observation into a *formula*.

# Backup: Two-Dataset Validation

Long-Range Dependency Gap: Classical vs. Neural LM



Gap is consistent across WikiText-2 and text8 (enwiki) — result is dataset-agnostic.